

HW 1.3a

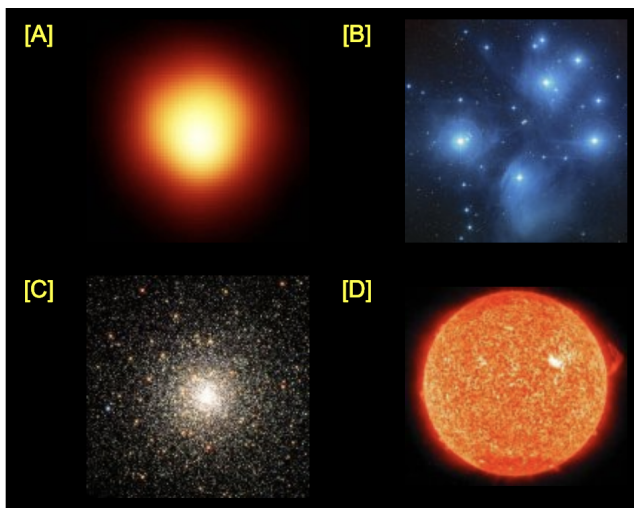
Name(s): _____

1. The sun is shrunk down to the size of a marble (1.4 cm radius).

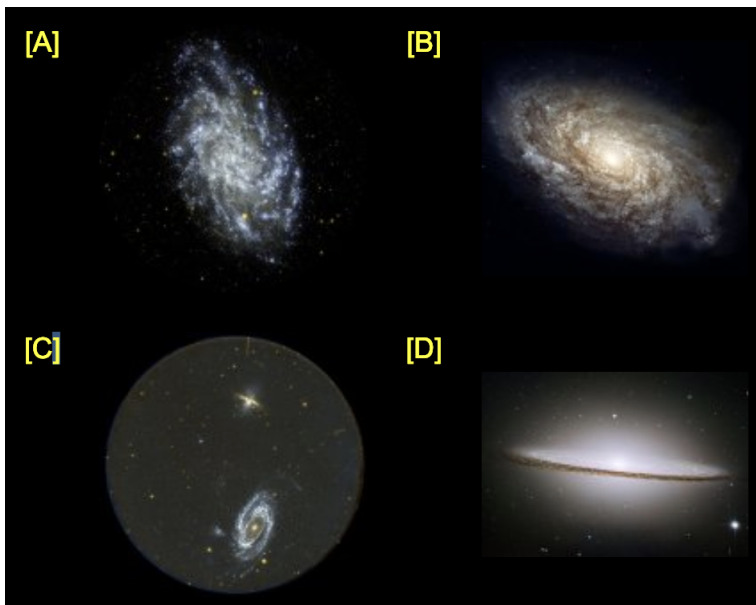
This represents a shrinking of _____ times.

Object	Actual Distance from Sun	Scaled Distance from Sun
1. Terra		
2. Pluto		
3. Proxima Centauri		
4. Center of Milky Way	160 kly	
5. Andromeda Galaxy	2 Mly	

2. If Neptune lies 4.2 light-hours from the Sun, and light travels at 300,000 kilometers per second, how many kilometers separate Neptune and the Sun?
3. What type of objects are shown here?



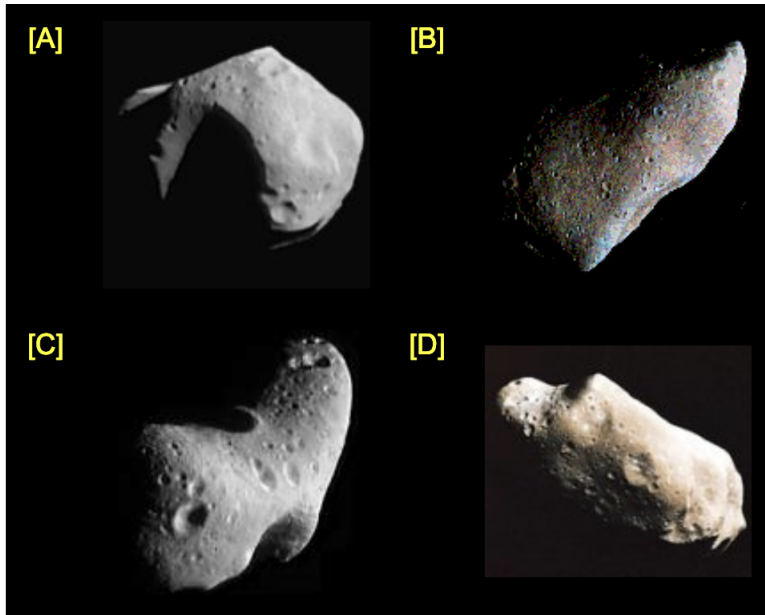
4. The diameter of the Earth is 12,800 kilometers, and the distance between the Earth and Sun is 150,000,000 kilometers. If you used a volleyball with an 8-inch diameter to represent the Earth, how far would you need to place it from your model Sun in a scale model of the solar system?
5. If it takes the Earth a full year to travel once around the Sun, covering a path which is almost one billion kilometers long, how long would it take a photon to cover the same distance, in units of minutes?
6. Planet Mars lies 78 million kilometers away from Earth at its closest approach, how long would it take for a radio signal sent from one of the NASA rovers exploring the surface of Mars to travel to us here on Earth, in units of minutes?
7. The Earth is _____light-minutes away from the Sun.
8. What kinds of objects are shown?



9. How long does it take the Earth to complete a full orbit around the Sun, in units of seconds?
10. If the center of our galaxy lies 27,000 light-years from the Sun, and light travels at 300,000 kilometers per second, how long does it take for light emitted from the galactic center to reach us, in units of years?
11. How many times more massive is the Sun than the Earth?
12. Exactly how many times larger is the Diameter of the Sun than the Diameter of the Earth?
13. If a light-year is the distance that light travels in a year at light speed, then by analogy a light-second is _____.
14. If it takes the Sun 240 million years to orbit once around the center of the Milky Way galaxy, were there any human beings around the last time we passed through this neighborhood?
15. Rank from smallest to largest: Earth Moon Jupiter Saturn Neptune Mars Sun

16. The speed of light is a constant. This means that if you compare the light coming from a flashlight held by a night watchman with the light coming from one held by a train engineer (rushing by on the midnight train), the light will _____.

17. What type of objects are shown here?



18. One of the first television transmissions from Earth to leak out into space showed the 1936 Berlin Olympic Games. In the year 2006, a star _____ away from us would have a chance to receive this signal (if it were loud enough).